

HUNTSVILLE, TX, USA

WMO: 722447

Lat: 30.744N

Lon: 95.586W

Elev: 366

StdP: 14.50

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 53903

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.0	31.3	14.6	11.7	38.9	18.6	14.2	42.0	19.8	63.3	18.2	61.1	3.4	340	0.371

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	20.0	98.9	74.6	96.9	75.2	94.7	75.3	78.5	88.6	78.1	88.1	77.5	87.4	5.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.3	139.2	81.7	75.4	135.1	81.2	75.1	133.4	81.1	42.2	88.9	41.6	88.0	41.1	87.7	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
17.1	15.2	12.7	DB	22.2	101.5	4.1	3.2	19.2	103.8	16.8	105.7	14.5	107.5	11.5	109.8
			WB	20.2	80.0	3.9	0.9	17.4	80.6	15.1	81.1	12.9	81.5	10.1	82.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	68.9	51.0	54.9	62.0	69.0	76.1	82.1	84.1	85.0	79.3	70.0	60.0	52.5
	DBStd	14.21	10.32	10.42	8.94	7.11	5.67	3.56	3.00	3.47	5.29	7.90	9.53	9.75
	HDD50	303	114	63	15	1	0	0	0	0	0	1	23	86
	HDD65	1618	448	306	165	43	4	0	0	0	1	43	204	404
	CDD50	7204	145	201	385	570	808	963	1058	1085	880	621	324	164
	CDD65	3042	13	24	70	162	347	513	593	620	430	198	55	17
	CDH74	31793	37	97	373	1174	3221	5673	7071	7664	4399	1732	297	55
	CDH80	14495	0	9	44	274	1224	2676	3551	4102	2010	566	38	1
(14) Wind	WSAvg	5.2	6.1	6.4	6.2	6.1	5.5	4.9	4.8	4.3	3.3	4.2	5.0	5.6
(15) Precipitation	PrecAvg	45.0	4.2	3.0	3.5	3.4	4.1	4.6	3.0	3.1	4.2	4.5	4.2	4.1
	PrecMax	61.0	11.5	7.2	7.0	9.3	12.8	15.2	7.0	7.5	9.5	18.1	14.2	8.4
	PrecMin	23.8	0.7	0.3	0.5	0.7	0.0	0.5	0.5	0.7	1.2	0.3	0.2	1.4
	PrecStd	9.4	2.5	1.9	1.7	2.1	3.0	3.2	1.9	2.2	1.9	4.1	3.1	1.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	78.4	81.3	83.9	89.3	94.1	99.6	99.9	102.4	99.7	91.4	83.6	79.2
		MCWB	65.1	64.4	67.2	71.9	73.8	73.9	74.3	75.1	72.5	72.7	70.8	68.7
	2%	DB	74.3	77.0	81.0	85.5	91.1	96.0	97.7	99.3	95.2	89.0	80.9	75.0
		MCWB	63.7	65.0	66.9	69.9	73.9	74.6	74.7	75.3	74.4	73.0	68.8	67.0
	5%	DB	71.7	73.4	78.6	82.9	89.4	93.2	95.4	97.2	92.5	86.2	77.6	72.3
		MCWB	63.9	64.3	66.0	68.9	73.6	75.3	75.3	75.7	73.6	70.9	66.6	65.8
	10%	DB	68.2	70.6	75.2	81.0	87.1	91.3	93.2	95.2	90.2	83.0	74.5	69.7
		MCWB	60.9	63.4	64.1	67.9	72.6	75.4	75.7	75.6	73.2	69.6	66.7	64.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.8	69.3	70.9	75.2	77.9	79.5	79.2	79.3	78.5	77.3	73.2	71.0
		MCDB	73.5	75.2	78.9	83.3	88.2	89.7	89.5	90.3	86.7	85.5	78.8	75.3
	2%	WB	67.0	68.1	69.5	73.2	76.5	78.3	78.3	78.4	77.7	75.9	71.5	69.0
		MCDB	72.1	73.7	76.3	80.9	86.0	88.0	88.4	89.3	86.0	82.9	76.9	73.5
	5%	WB	65.4	66.7	68.5	71.9	75.4	77.5	77.8	77.9	76.9	74.7	70.2	67.3
		MCDB	70.4	71.8	74.8	79.1	84.1	86.9	87.8	88.7	85.2	81.4	75.4	71.5
	10%	WB	62.4	64.9	67.2	70.8	74.5	76.9	77.3	77.3	76.1	73.3	68.3	64.8
		MCDB	66.9	69.5	73.0	77.7	83.0	86.1	87.1	88.1	84.3	80.0	73.6	68.4
(35) Mean Daily Temperature Range	5% DB	MDBR	19.2	18.8	19.8	19.7	18.9	17.8	18.4	20.0	20.0	21.1	20.0	19.2
		MCDBR	20.3	19.6	19.6	20.0	19.6	19.9	21.1	22.2	22.1	21.1	20.2	20.4
		MCWBR	12.7	11.4	9.2	8.4	6.4	5.2	4.7	4.7	6.2	7.9	10.7	14.0
	5% WB	MCDBR	17.9	17.3	17.1	16.3	17.4	16.6	17.8	19.7	17.5	17.2	16.7	18.6
		MCWBR	13.6	12.7	10.7	8.8	7.2	5.4	5.0	4.8	5.8	8.0	11.1	15.6
(40) Clear-Sky Solar Irradiance	taub		0.327	0.349	0.367	0.417	0.469	0.460	0.478	0.469	0.437	0.370	0.367	0.332
	taud		2.516	2.467	2.416	2.279	2.171	2.248	2.229	2.255	2.300	2.473	2.424	2.512
	Ebn at Noon		287	290	291	279	263	264	259	260	265	279	269	279
	Edh at Noon		28	32	36	43	48	44	45	43	40	31	30	27
(44) All-Sky Solar Radiation	RadAvg		904	1011	1384	1692	1843	1974	1997	1894	1609	1340	996	788
	RadStd		86	189	128	110	139	161	150	134	148	179	108	99

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+171
(47) Regional (39 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+194	+167

Nomenclature: See separate page